

Tanmay Singh

Delhi, India | +91 9660413894 | tanmay.work10@gmail.com | [LinkedIn](#) | [Github](#)

Data Engineer with experience designing and scaling ETL pipelines, relational schemas, and cloud data infrastructure for AI-driven products and enterprise financial systems. Skilled in building reliable, high-throughput data platforms across Airflow, BigQuery, PostgreSQL, and Supabase, with a track record of cutting pipeline runtime and processing 10M+ records monthly at production scale.

EDUCATION

Arizona State University — Master of Science in Data Science, Analytics and Engineering
Manipal University, Jaipur — Bachelor of Technology in Computer Science and Engineering

EXPERIENCE

Vault IQ

Dec 2025 – May 2026

Data Engineer (AI and Backend Systems)

Remote

- Designed and implemented relational data schemas and storage systems for user, brand, product, and social data using Supabase, enabling structured data flow across 4+ product modules
- Architected a full-stack, AI-powered brand development platform integrating multi-model LLM and generative AI systems (OpenAI, Gemini, Flux Pro) to automate content generation and product visualization pipelines
- Built a membership platform and dashboard backend to serve brand portfolio, product catalog, and revenue metrics data in real time, alongside UI/UX improvements across core surfaces
- Developed and deployed an AI-powered chatbot with a custom knowledge-base integration layer, supporting contextual querying and FAQ automation in production
- Engineered an AI photoshoot generation system using reasoning-based prompt orchestration, integrating LLM planning with image-generation APIs while preserving product fidelity
- Integrated 6+ external services and APIs to automate data ingestion, content generation, and cross-module functionality across the platform

Capgemini Pvt Ltd

Oct 2020 – Mar 2023

Data Engineer – HSBC UK (Application Data Management)

United Kingdom

- Designed and orchestrated scalable ETL pipelines using Apache Airflow (DAG-based workflows) to support batch processing for HSBC's customer credit data systems
- Managed and processed 10M+ customer records monthly across multiple environments using Google BigQuery and PostgreSQL, ensuring data consistency, integrity, and availability for downstream applications
- Developed data transformation and validation workflows using Python and PySpark, implementing data quality checks aligned with business-critical credit risk models
- Optimized BigQuery queries, storage structures, and retrieval performance, reducing batch pipeline runtime by 40% and improving analytical efficiency
- Built monitoring into pipeline runs to catch data anomalies pre-emptively, improving reliability of production-grade credit data systems
- Collaborated within cross-functional teams to maintain and enhance production-grade data systems, ensuring stability and timely delivery

PROJECTS

Reducing Carbon Emissions in Mesa, AZ | Urban Transit and Environmental Analysis

- Built a data pipeline combining structured and unstructured transit, temperature, and demographic datasets to power geospatial analysis in QGIS
- Modeled population density and transit-gap layers across city tracts to engineer 5 new bus routes and 45 stops with buffer and intersection analysis
- Integrated city transportation budget data into the pipeline to generate preliminary cost estimates for route expansion and operations
- Modeled ridership growth scenarios (5–15% uptake) end-to-end, projecting average annual CO2 savings of 8,000 metric tons

Weather-Based Solar Power Forecasting

- Built a data pipeline to source and merge time-series datasets from the NASA POWER API with solar panel performance data for model training
- Built and evaluated 3 regression models (Linear, Polynomial, SVR with RFECV) using scikit-learn pipelines and automated feature selection, achieving MAE of 10–15 kWh/day
- Automated evaluation reporting with Matplotlib visualizations comparing actual vs. predicted output and model diagnostics

SKILLS

Languages & Frameworks: Python, SQL, PySpark

Data Platforms & Warehousing: PostgreSQL, Google BigQuery, Supabase, Data Modeling, Schema Design

Pipelines & Orchestration: Apache Airflow (DAG-based ETL), Batch Processing, Data Validation & Quality Checks

Cloud & Infrastructure: AWS, Google Cloud, Docker, Git

Other: REST API Integration, Prompt Engineering, LLM/Generative AI Integration, Cross-functional Collaboration